INDIA INNOVATES 2026



Team Name:
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From Graphic Era Hill University, Dehradun

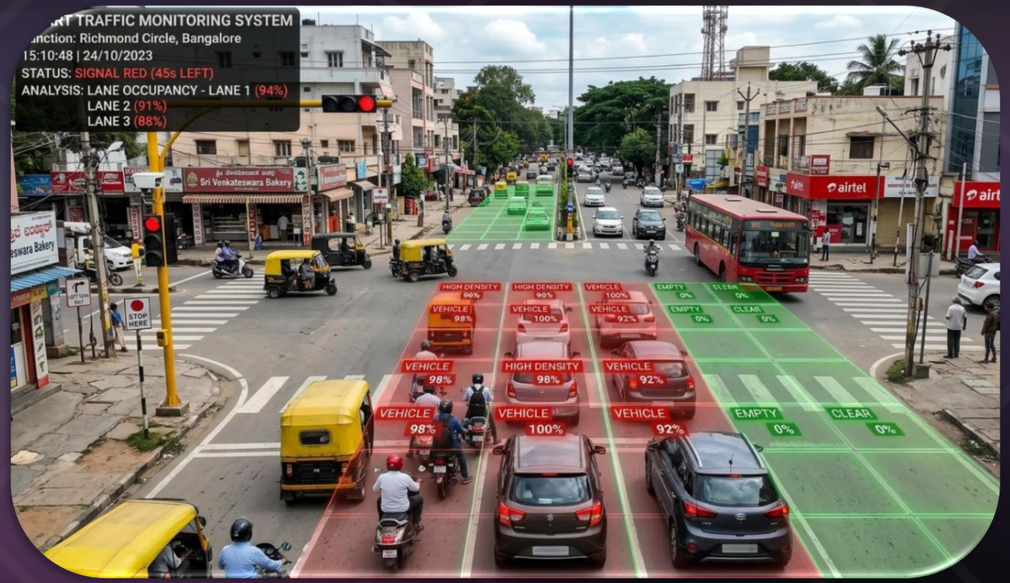
PROBLEM STATEMENT

Given:
Create an intelligent traffic management system that uses real-time computer vision to dynamically adjust signal timings based on live traffic density. The system must also enable an AI-powered green corridor feature to prioritize and clear routes for emergency vehicles such as ambulances and fire services.

Explanation:
Modern metropolitan cities are experiencing rapidly increasing traffic congestion due to urbanization, population growth, and rising vehicle ownership. Cities such as **Delhi** frequently face severe traffic bottlenecks during peak hours, resulting in longer travel times, higher fuel consumption, increased pollution, and delayed emergency response services.
Current traffic infrastructure primarily relies on **static signal timings**, where traffic lights operate on predefined schedules regardless of real-time traffic conditions. This approach often leads to inefficient traffic flow because it cannot dynamically adapt to varying congestion levels across different lanes and intersections.
Key limitations of existing systems include:
• Lack of **real-time traffic awareness** at intersections
• Poor coordination between neighboring traffic signals
• No automated mechanism to **prioritize emergency vehicles** such as ambulances or fire services
• Delayed detection of accidents and road blockages
• Inefficient traffic routing leading to congestion and fuel wastage
These inefficiencies affect daily commuters and can become **life-threatening when emergency vehicles are delayed in traffic**.

Therefore, there is a critical need for an **intelligent, adaptive, and scalable traffic management system** capable of analyzing live traffic conditions, dynamically optimizing signal timings, and enabling **priority corridors for emergency services**.

SOLUTION



To address the limitations of traditional traffic infrastructure, we propose a **Dynamic AI Traffic Flow Optimizer and Emergency Grid**. The system integrates computer vision, deep learning, and reinforcement learning to monitor real-time traffic conditions and dynamically control traffic signals across intersections. Traffic camera feeds are analyzed using computer vision to estimate lane congestion through a **grid-based road occupancy model**, which measures how much road space is occupied rather than simply counting vehicles. The system then uses a combination of **Deep Learning for traffic understanding** and **Reinforcement Learning for adaptive signal control**, allowing signals to continuously improve decision-making by learning from real traffic patterns over time.

Key Components

1. Real-Time Traffic Monitoring & Grid Density Analysis

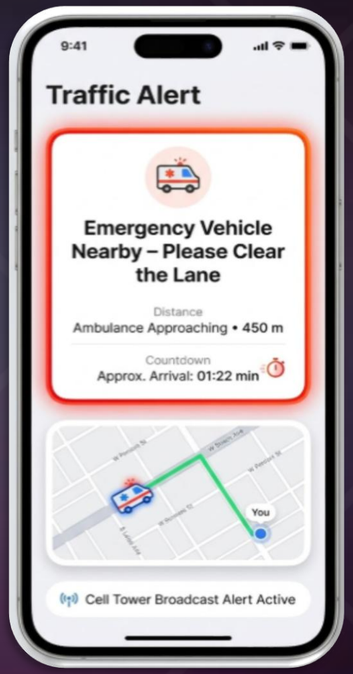
Traffic cameras capture live intersection footage and vehicles are detected using **YOLOv8**. Each lane is divided into grid cells to calculate **road occupancy**, providing a more accurate traffic intensity score even in mixed traffic conditions involving cars, bikes, buses, and auto-rickshaws.

2. AI-Based Dynamic Signal Control

Deep learning models analyze traffic density, queue lengths, and vehicle movement patterns to understand congestion levels. A reinforcement learning controller (such as **Deep Q-Network**) acts as a **dynamic policy engine**, continuously optimizing signal timings and learning from traffic behavior to improve traffic flow over time.

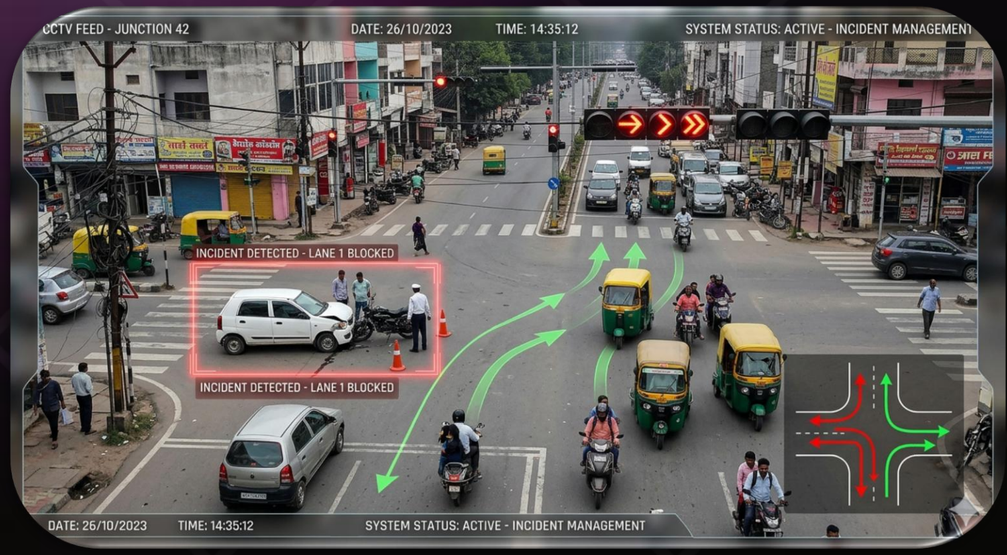
SOLUTION

3. AI-Powered Emergency Response System
The system ensures faster emergency response through multiple coordinated mechanisms:
• **Green Corridor Creation** – Signals along the emergency route automatically synchronize to allow uninterrupted movement for ambulances and fire services.
• **Citizen Alerts** – Nearby drivers receive emergency notifications through **Cell Broadcast Service**, requesting them to clear lanes.
• **Emergency Navigation App** – A dedicated mobile application helps emergency vehicles identify the fastest available route using real-time traffic intelligence.





4. Accident Detection & Smart Rerouting
Computer vision models detect abnormal traffic patterns such as stalled vehicles or sudden congestion spikes. When an accident is detected through the trained model, the system highlights alternative routes using **dynamic red direction indicators**, guiding vehicles away from the affected area and reducing traffic buildup.



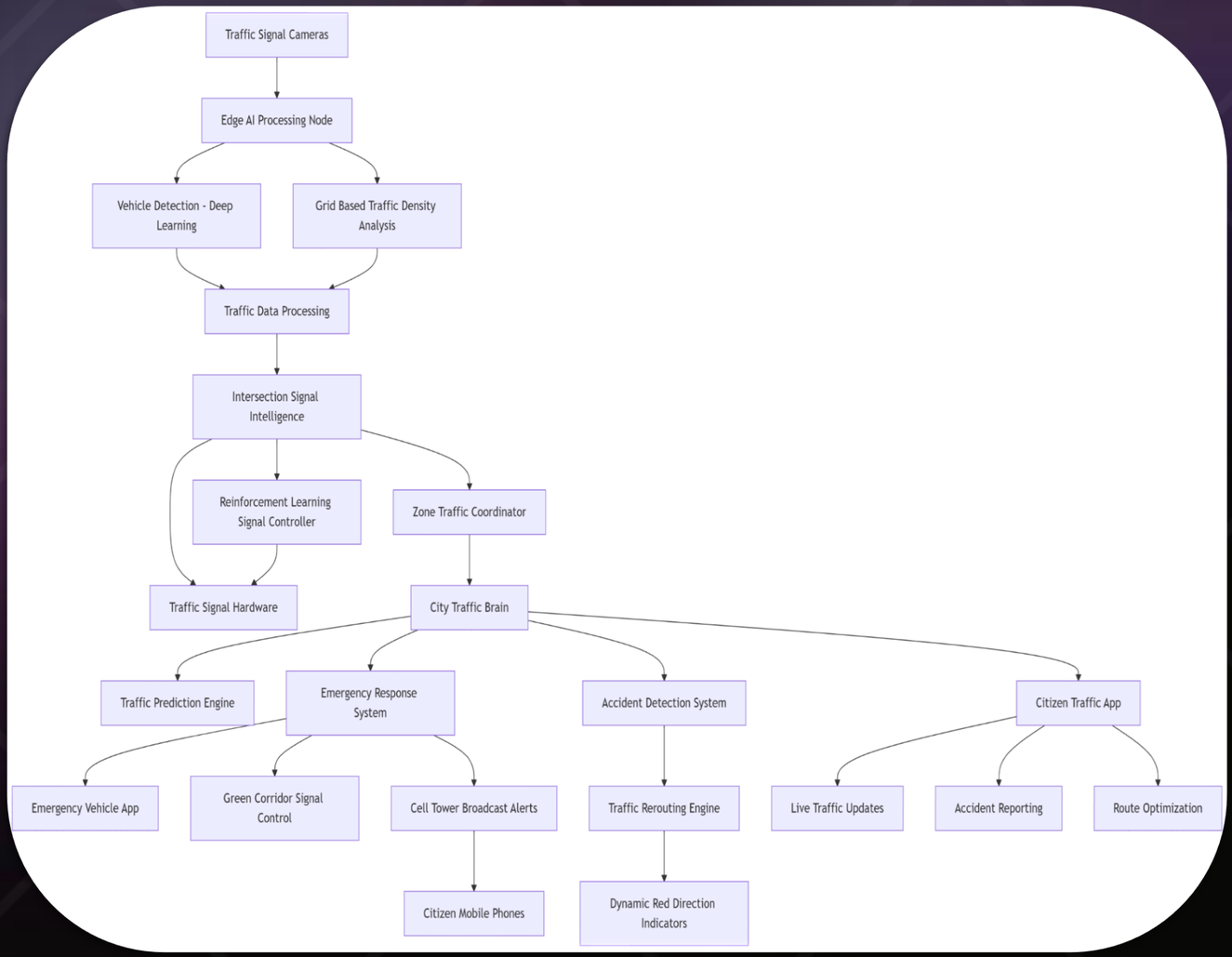
5. Citizen Traffic Intelligence App
A companion mobile application allows users to:
• receive emergency vehicle alerts
• view live congestion information
• report accidents or road blockages
• access optimized fastest routes suggestions instead of shortest
This creates a **crowd-assisted traffic intelligence network** that improves system awareness and responsiveness.

ARCHITECTURE

System Overview

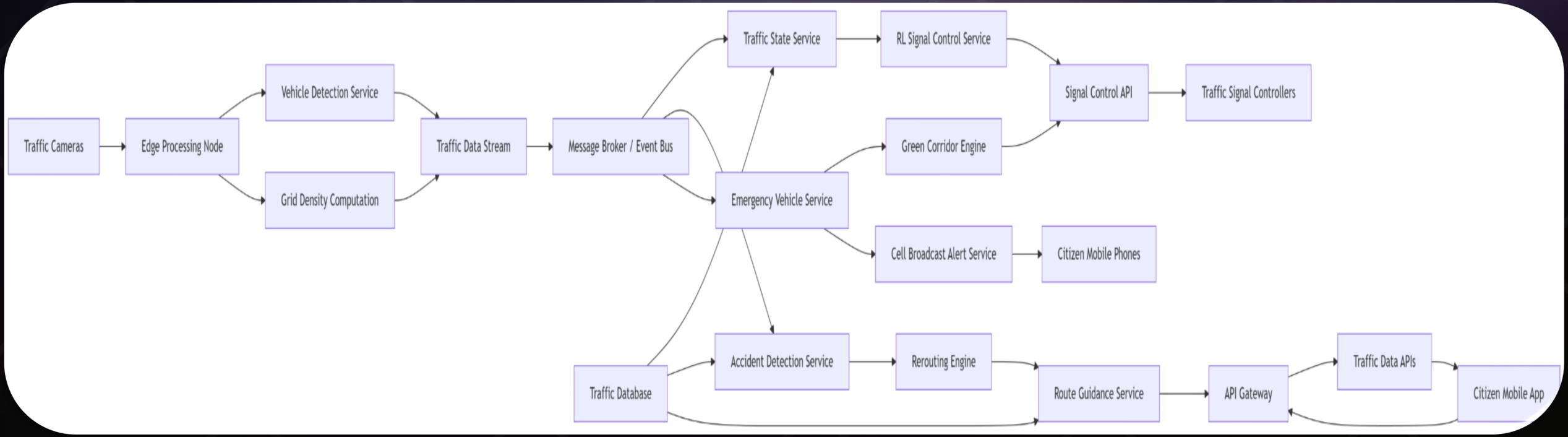
- The system follows a **hierarchical distributed architecture** designed for scalable city-wide deployment. Intelligence is distributed across multiple layers to ensure real-time responsiveness, reliability, and efficient traffic coordination.
- At the **edge layer**, traffic cameras capture live road activity. Edge AI modules process these feeds to detect vehicles and estimate traffic density using computer vision.
- The processed traffic data is sent to **intersection-level AI controllers**, where reinforcement learning models dynamically optimize signal timings based on congestion levels and queue lengths.
- Multiple intersections are coordinated by a **zone traffic controller**, which synchronizes signals to improve traffic flow and create smooth green corridors.
- At the top level, the **city traffic intelligence layer** aggregates data from all zones to monitor congestion, predict traffic patterns, and manage emergency vehicle routing.
- An **Emergency Grid system** can override normal traffic control to automatically create green corridors for ambulances and fire services.

System Flow



ARCHITECTURE

System Design



TECHNOLOGY USED

Computer Vision & Traffic Detection
- **YOLOv8** – vehicle detection from traffic camera feeds
- **OpenCV** – video processing and lane region extraction
- Grid-based lane segmentation for traffic density estimation

AI & Traffic Optimization
- **Deep Q-Network** – dynamic signal control policy
- **Proximal Policy Optimization** – reinforcement learning optimization
- Deep learning models for traffic pattern analysis

Simulation & Model Training
- **SUMO** – traffic environment simulation for backend dashboard and RL training

Backend Infrastructure
- Python / FastAPI or Node.js for traffic APIs
- Real-time traffic data processing and signal control services

Mobile & Communication Systems
- **Flutter** – cross-platform citizen and emergency vehicle mobile apps
- **Cell Broadcast Service** – emergency alerts to nearby drivers
- Navigation integration similar to **Google Maps**

FEATURE / USP

Grid-Based Traffic Density Analysis
Instead of simple vehicle counting, the system divides each lane into **rectangular grids** and measures **vehicle vs empty road occupancy**. This enables far more accurate congestion estimation in **mixed Indian traffic conditions** with cars, bikes, buses, and auto-rickshaws.

AI-Powered Traffic Signal Intelligence
The traffic signal system uses **Deep Learning for traffic understanding** and **Reinforcement Learning for adaptive signal policies**. Over time, the model learns traffic patterns and continuously improves signal timing efficiency.

Emergency Grid & Fastest Route Optimization
Emergency vehicles use a **dedicated navigation app** that calculates the **fastest route based on real-time congestion**, not just shortest distance. The system automatically synchronizes signals to create a **green corridor across intersections**.

Cell Tower Emergency Alerts
Nearby drivers receive **instant mobile alerts through cell broadcast systems**, notifying them of approaching emergency vehicles so they can **clear lanes before the vehicle arrives**.

Accident Detection & Smart Rerouting
Computer vision detects accidents or unusual traffic stoppages. Traffic signals then display **dynamic red directional arrows** guiding vehicles toward **alternate free routes**.

Citizen Traffic App
A companion mobile app provides:
- real-time congestion updates
- emergency vehicle alerts
- accident reporting
- optimized route guidance

Scalable Smart City Deployment
The system integrates with **existing traffic cameras and signal infrastructure**, making it feasible for **large-scale city-wide deployment** with minimal additional hardware.

REFERENCES / LINKS

The following research papers and resources provided technical guidance for the development of this solution.
Reinforcement Learning for Traffic Control:

• Traffic Signal Control With Deep Reinforcement Learning
https://ieeexplore.ieee.org/document/8054579
• Deep Reinforcement Learning for Traffic Light Control in Vehicular Networks
https://ieeexplore.ieee.org/document/9098073
• PressLight: Learning Max Pressure Control to Coordinate Traffic Signals
https://dl.acm.org/doi/10.1145/3292500.3330949

Traffic Simulation:

• SUMO
https://www.eclipse.org/sumo/

Computer Vision:

• YOLOv8
https://github.com/ultralytics/ultralytics

THANK YOU